9. 12. 2025

Systems Concepts & IS Environment

system → particular objective fulfil
info → processed data
data → representation of facts
No manipulation
Interest group এর মধ্যে যা we know
processed data → particular obj attain করতে some data/entity deleted
knowledge → decision making এ needed
55 desks
50 students + 5 surplus → অন্য class এ ভাঙা desks আবশ্যক replace করা possible

This chap → at least 15 marks
(elementary topics → *)

system → I/P (info) → Processor → O/P (decisions)
No control over given inputs.
(from environment → takes I/P) → Process action → O/P (given to environment)

What is system? → keywords (VVI *)
components → I/P, O/P > environment
↳ multiple comp → interdependent / interconnected
↳ system এর obj fulfil
example → explain

characteristics of system

1. Org → organogram.
   hierarchical → top-down
2. Interaction → component to component interaction
3. Interdependence → attendance no. + CT mid no. + Final no. (dependent)
4. Integration → sossiorial, theory → result combine
5. central obj → Result process

→ 405 & 415 not interdependent.

30.12.2025

system Models :

components
Module to Module connect
Flow → Unidirectional / bidirectional.

(a) Scenario → scematic diagram draw.

* Flow diagram → কোন component কাকে trigger, কারার outcome for input.

Perct → Program evaluation & review technique.
start, end time, কে কাকে ignite করছে।
কাজের flow define করা হচ্ছে।
কোন task কখন execute হচ্ছে, নিজে কার উপর dependent, কাকে ignite করছে (gives ip to another)

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(Q) Flow + gantt
" + scematic.

* Gantt chart → resource allocation
progress, timeline, dependency.

Scematic, flow, gantt → sequential

Types of system :

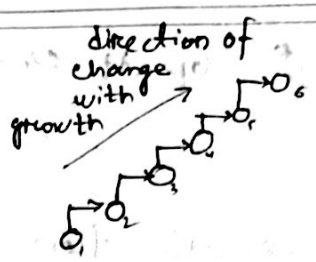
physical → computer, CPU
abstract → S/W system, DB (result processing system)

open, closed
deterministic → ATM machine
Probabilistic → weather forecast
Man-made (IS) → Uniplex (particular obj. accomplish করতে)

Imp **

Nolan stage Model

1. Initiation → discussion about problem
2. Contagion → আলাদা modules
3. control → 1 stage এ all module
4. Integration → integrated modules (ERP)
5. Data administration → data backup & manage daily. ensures the functionalities of modules. If system fails, data should be fetched from backup.
6. Maturity



সব org. এর সাথে applicable.

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20. 01. 2026
Zoom Class
Prof. Azhar

System Analyst

* bridge betw^n users & developers
* analyzes, designs & improves IS.

Role → studies existing system,
user reqs ?
designs system sol^{ns}
coordinates with stakeholders.

Qualification → B.S.C in CSE/ IT/IS
MSC, Prof. certification (optionals).

Personal Qualification → Problem solving skills
comm. skills
Teamwork & leadership
Adaptability & Ethics

Roles → consultant.

SDLC

User requirement, user friendly UI, feedback.

* Behavioral issues ** (examples + meaning of the issues).
* managing the "

Think, Pair & Share

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The problem is user resistance due to poor req. analysis & change management & the system analyst is primarily responsible.

Sol^n → Provide proper user training & involve users in system design improvement.

User should always be involved throughout SDLC. Their requirements are the building blocks to build a system.

System analyst → did not ~~share~~ encourage user participation & feedback.

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19.2.2026

IS Design & Dev.

Chap 4

System Planning & the Init Investigation

* Planning in system analysis

→ কার, কাদের দিয়ে, expertise কি, কখন কাজ, কি কি কাজ

কি, কখন, কিভাবে, কাদের দিয়ে কাজ করাবো
(what) (when) (how) (who)

↳ 4 keywords (**)
(identify)

→ time restriction ?

less than 2 yrs → short
2-5 → medium
5+ → long / mega / range project.

→ focus (strategic → next কোনো program start করার জন্য যেসব decisions,

managerial → guideline
(specific requirements)
(decision execute করতে যা যা করা লাগবে)

operational → programmers, developers.

* Init Investigation activities :

i) Needs identification

req. engineering → particular কাজের জন্য কোন model follow করা লাগবে।

autism → ethnography. study
× interview.

justification for using a model.

① Real prob or symptoms থেকে সমস্যা

② asking
getting info
prototyping

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2) Determine user's Info req.

meaningless req → বাদ

user এর req. → all must be stated clearly or technically.

3) Problem def^n (1 statement)

- subques. } using all stated reqs).
- challenge

4) Bg. analysis → so far কি কি করা হয়েছে।

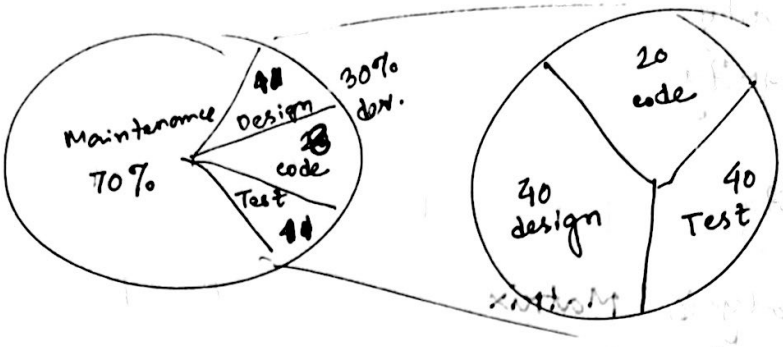
5) Fact Finding → real fact কি কি ঘটনা

6) " Analysis

7) Determination of feasibility.

2.4.2026

Chap 13
SW Maintenance



dev. cost

Challenges :

1. Technical expert group ~~ता~~ continuation
2. new project build is better than updating requirement of existing project
3. Types → corrective
   Adaptive
   Perfective
   Preventive (threat ~~tackel~~ ~~A~~ affected ~~A~~ 2,3-)

Reduce M. cost

1. M. Mgt Audit
2. S/W Sys. "
3. S/W Modification

chap 16
{ System Security }

system security
" Integrity
Privacy
confidentiality

*** Risk Analysis Matrix

(Q) Scenario about a system → R.A. Matrix
(*** expected as ans).

Probable risk factors
↓
Impacts → cost of Risk → Prob.
(high, Mid, Low)
0.5 0.3 0.2

↓

$$\frac{\text{cost} \times \text{Prob}}{1000}$$

→ max → higher priority to solve the risk

control measures.
↳ 4

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